

Design → Demo → Deploy

An AI intake & referral agent for Workers' Comp DME — live today, deployable in your environment in 8–12 weeks. A hands-on workshop to prove the design, the demo, and the rollout.

1 Day

WORKSHOP

Agent 1

LIVE NOW

\$0.032

PER REFERRAL

24

CMS RULES

Make intake & referral management an advantage

Today, DME referral intake is manual, slow, and error-prone — staff rekey faxes, chase missing fields, and miscode claims. The agent turns it into autonomous, clean, route-ready episodes — with a human in the loop at every flagged exception.



The cost of manual intake

Hours of rekeying per referral · coding errors that trigger denials · missing auth refs that stall episodes · no audit trail · can't scale with volume.

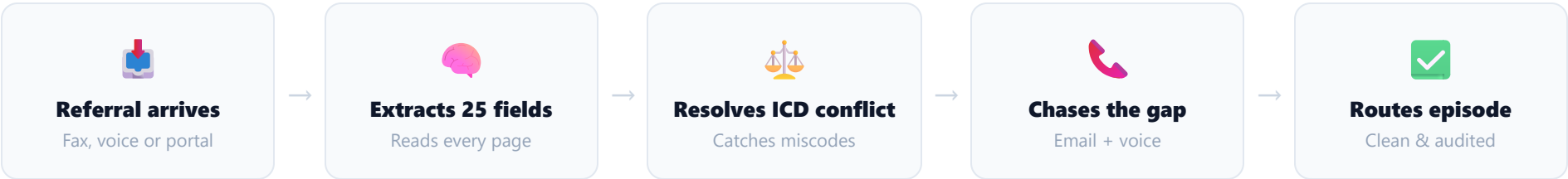
The agent advantage

Reads every page, extracts 25 fields, validates against CMS rules, chases gaps by email & voice, and routes a clean episode — in seconds, at \$0.032 each, fully audited.

The principle: autonomous where it's confident, human where it isn't. The agent never guesses on a clinical or coding decision — it escalates.

What you'll see — on real referrals

Not a mock-up. A real Claude agent processes actual referral PDFs end-to-end, on screen, on your representative data.



Watch for #1

The agent flags an ICD conflict (M23.611 vs S83.209A) and explains *why*, citing the document.

Watch for #2

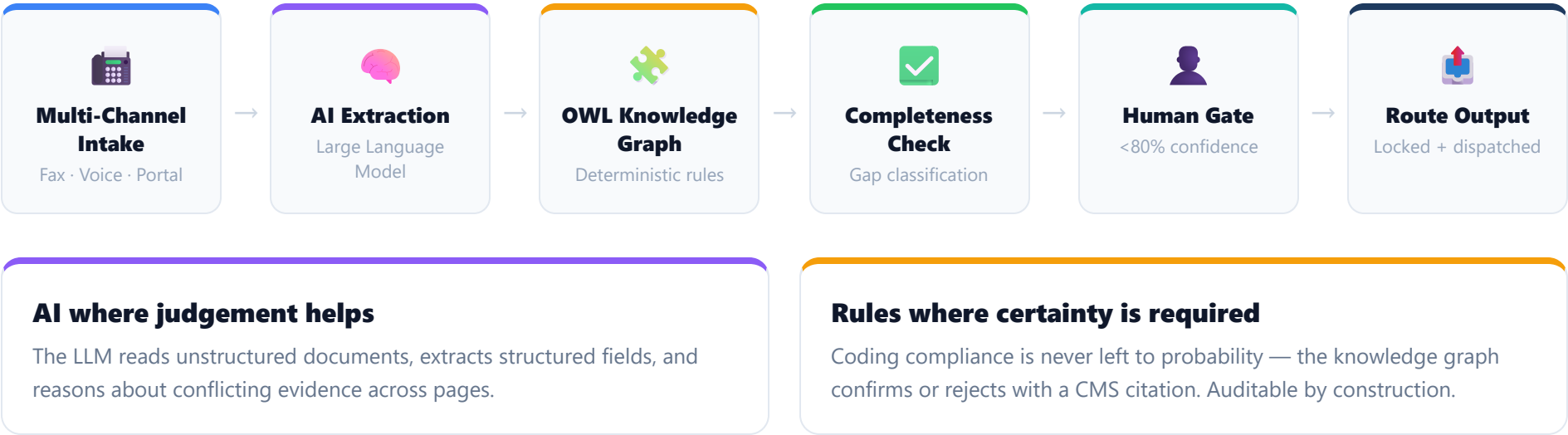
A missing auth ref is caught as a hard block — the episode is held, outreach is drafted to the adjuster by name.

Try it yourself

Scan the QR → open the live demo on your phone → run a referral and try to “stump the agent.”

The big picture — six layers, one pipeline

Every referral, regardless of channel, flows through the same six-layer pipeline. AI does the reading and reasoning; a deterministic rule engine does the compliance; a human decides the close calls.



How the agent works — step by step

- 1 Channel Intake**
Ingests fax PDFs (counts every page) or answers the live voice line — 14 fields captured in ~107 seconds. Both channels feed one queue.
- 2 AI Extraction — 25 structured fields**
Reads all documents at once: patient, claim, ICD-10, HCPCS, physician NPI, adjuster, address, auth ref. Cross-references documents and scores its own confidence 0–100%.
- 3 OWL Knowledge Graph — 24 CMS rules**
Every code validated against deterministic rules sourced from CMS guidelines. Confirms, rejects, flags or escalates — each with its citation. Runs in <150 ms.
- 4 Completeness Check**
Classifies gaps by criticality: hard blocks (auth ref, appt window) hold the episode; required gaps (transport, language) collected in parallel; clean episodes route untouched.
- 5 Human Review Gate**
Below 80% confidence the agent stops — routes to a specialist with full reasoning, source docs, one-click approve/override, logged with reviewer identity.
- 6 Route Output**
Locks the 25-field episode record, saves the full audit trail, triggers vendor dispatch with the right code, equipment and patient.

Multi-channel in, autonomous outreach out

The agent meets referrals where they are — and chases what’s missing without a human lifting a finger.



Fax

Ingests all PDFs — referral form, clinical notes, prescription — and counts every page before forming an opinion.



Voice (inbound)

Answers the intake line 24/7, runs a structured conversation, captures 14 fields live — 107s average.



Portal upload

Accepts portal submissions into the same queue — same process, same output.



Email outreach

Drafts a specific email to the adjuster *by name* — lists exact missing fields, cites the claim number, sets a 24-hour deadline. Logged on send.



Outbound voice call

No reply in 24h? The agent calls — 84 seconds — confirms auth ref, collects the appointment window, updates the episode from the transcript.

Four-layer defense against hallucination

The single most important design choice: the agent is built so it cannot quietly guess on a clinical or coding decision.



1 • Document cross-reference

Compares ICD codes across every document — mismatches flagged with document-level reasoning.



2 • KG validation

Deterministic CMS rules confirm or reject each code — no probability on coding.



3 • Confidence threshold

The agent scores itself 0–100%. Below 80%, it does not proceed.



4 • Human gate

Low-confidence or conflicting cases route to a specialist for a binding decision.

Status priority — the guardrail that makes it deployable: REJECTED > ESCALATED > FLAGGED > VALIDATED. The safest outcome always wins — the agent never marks an episode “clean” while a live coding concern exists.

Every decision is logged & reviewable

When the agent escalates, the specialist sees *why* — not just a flag. When it acts, the reasoning survives audit.

The human review queue

- ICD conflicts below 80% → specialist queue, episode blocked from dispatch
- Reviewer sees the agent's full reasoning + source documents
- Approve or override in one click
- Decision logged with reviewer identity + timestamp

The audit trail (per episode)

- 25 structured fields + KG validation results
- Every rule that fired, with CMS citation
- ICD conflicts and how they were resolved
- Email body + call transcript
- Every confidence score, every decision

“When confidence falls below 80%, the agent stops. It doesn't guess on clinical decisions — it routes to a specialist with the full conflict detail.”

A deterministic rule engine — cited, not guessed

24 rules extracted from CMS ICD-10-CM Official Guidelines FY2026 and LCD/NCD documents. Every validation is deterministic, cited, and auditable — no AI probability on coding decisions.

24

RULES

7

CATEGORIES

100%

DETERMINISTIC

<150ms

PER EPISODE

Why OWL, not a flat rule list

OWL (W3C Web Ontology Language) enforces class hierarchies, property chains and disjointness — so the graph stays logically consistent as rules are added. A reasoner can infer new facts automatically.

Built from the source of truth

CMS ICD-10-CM FY2026 (144 pages) + LCD L33803 (walkers/rollators) + NCDs → structured into `rules.json`, each rule citing its exact CMS section.

Seven rule domains • four possible actions

Post-Surgical Coding 6 rules — enforce S-codes (injury) over M-codes (degenerative) after surgery.	Pain Classification 3 rules — postprocedural G89.18 vs trauma G89.11; sequencing.	Laterality 2 rules — right/left mismatches across documents; escalate conflicts.	HCPCS Coverage 4 rules — confirm equipment codes covered for the diagnosis.
Equipment Spec 2 rules — bariatric thresholds (E0143 300lb vs E0149 400lb).	Workers Comp 3 rules — required WC fields; reject M-codes as primary on injury claims.	Authorization / Docs 4 rules — missing auth ref; 5-element medical-necessity check.	Every rule cites CMS e.g. §I.C.19.a, LCD L33803 — traceable to source.
CONFIRM Code validated — contributes to VALIDATED status.	REJECT Coding error identified — replacement + source given.	FLAG Concern needs review — may not block dispatch.	ESCALATE Conflicting evidence — human decision required.

Live example — the Holloway referral

James Holloway · ACL reconstruction · E0143 rollator · 285 lbs. The KG ran 24 rules in <100 ms — 5 fired.



DME-ICD-001 — REJECT

M23.611 (degenerative) on a post-surgical episode. Replace with S83.209A. Source §I.C.19.a.



DME-ICD-002 — CONFIRM

S83.209A correct post-ACL encounter code (7th char 'A'). Confidence 100%.



DME-ICD-007 — CONFIRM

E0143 rollator covered for S83.209A per LCD L33803.



DME-ICD-023 — FLAG

285 lbs — within 15 lbs of the 300 lb limit. Confirm bariatric spec before dispatch. Catches a costly wrong-equipment return.



DME-ICD-015 — FLAG

WC claim, auth ref blank — blocks billing. Triggers outreach to adjuster Linda Torres.

Final status: FLAGGED — 1 rejection (resolved by LLM self-correction), 2 confirmations, 2 flags. Full citation trail attached to the route output.

Fits your stack — no rip-and-replace

The agent connects to what you already run, three ways, and is built to slot into the wider operations chain.



API

Direct integration with your case-management / EHR / intake portal where APIs exist.



RPA

Robotic automation drives legacy systems that have no API — the agent operates the screens.



File

Batch file exchange (fax gateway, SFTP drops) for the simplest, lowest-touch start.

Connects to

Fax gateway · intake portal · case-management system · scheduling · billing · EHR — inbound and outbound.

Carrier rules live in the graph

Payer-specific rules are added to the knowledge graph as config — no code changes to support a new carrier.

PHI stays inside your boundary

Healthcare-grade from day one. This is the question your technology leader asks first — here is the answer.



HIPAA + BAA

HIPAA-compliant deployment. LLM-provider BAA available. PHI never leaves your boundary.



Access & audit

Role-based access, full audit logging of every decision, configurable data-retention policy.



Your environment

Containerized Python — runs on your infrastructure, your cloud, or ours. You receive source, logs, runbooks.

Bring this to the security review: data-boundary diagram, HIPAA/PHI checklist, draft BAA summary, retention policy — all on the table at the workshop.

Enterprise-ready operations

Everything a technology leader needs to say yes — and to run it confidently after go-live.



Monitoring

Live system health, AI quality metrics, throughput & cost dashboards.



Support

Dedicated support through pilot and go-live; runbooks for your team.



Scales

Demonstrated 40 → 1,000 → 3M referrals on the same architecture.



Improves

Learns your payers and patterns; review-queue feedback feeds the next cycle.

“Your team receives full source code, audit logs, and runbooks — with dedicated support through pilot and go-live.”

The economics — measured, not estimated

Real token consumption from 40 live referrals, extrapolated to production volume.

\$0.032

PER REFERRAL

40

LIVE REFERRALS MEASURED

1,000

EXTRAPOLATED

3M

AT FULL SCALE

What drives the cost

A few cents of LLM tokens per referral for extraction + reasoning. The knowledge graph runs locally — no per-call cost, no network round-trip.

Why it stays flat at scale

Cost is per-referral and linear — no per-seat licensing, no overtime, no rekeying. Volume spikes don't spike headcount.

Where the return comes from

We'll plug your monthly volume into the calculator live — these are the levers.



Time saved

Seconds of agent processing replace minutes-to-hours of manual rekeying and gap-chasing per referral.



Fewer denials

Coding errors caught before submission — M-code rejections, missing auth refs — mean fewer reworks and denials.



Throughput

Process peak volume without adding staff; clean episodes route with zero human touch.

The pitch in one line: at \$0.032 per referral, the cost of the agent is a rounding error against a single avoided denial or hour of staff time. We'll prove the payback on *your* numbers.

What we built • how we deploy • how it improves

01 — What we built

A modular agent network: AI extraction, OWL knowledge graph (24 CMS rules), four-layer defense, email + voice outreach, full audit trail. **75 referrals in 60 seconds at \$0.032 each.**

02 — How we deploy it

Containerized Python in your environment. API / RPA / file integration — no rip-and-replace. BAA + HIPAA config. **8–12 weeks from kickoff to go-live.**

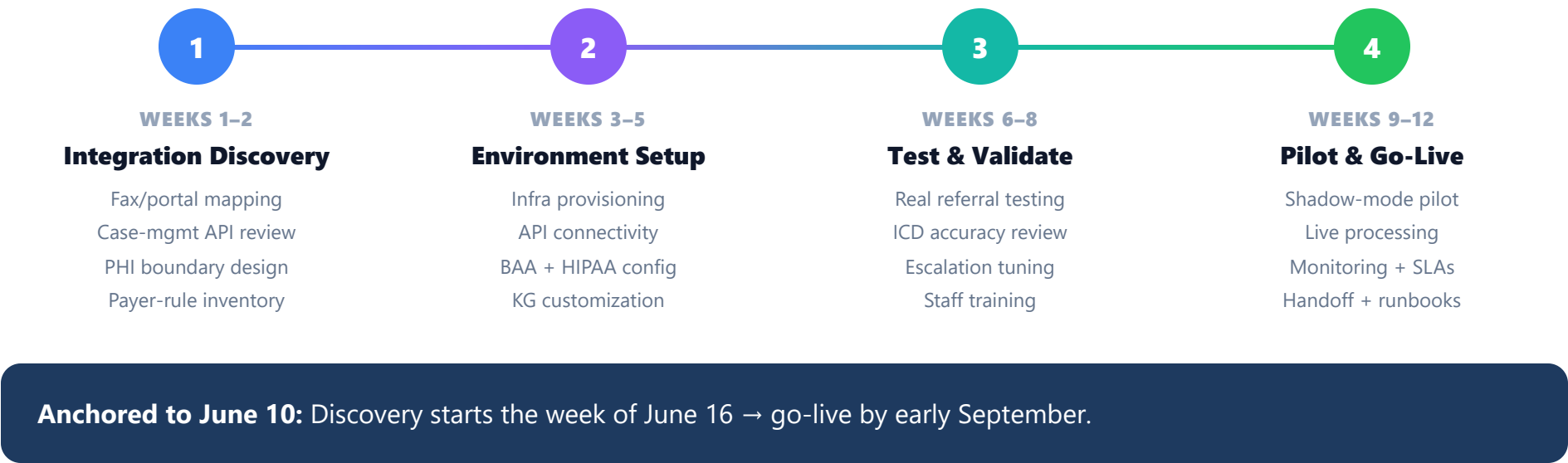
03 — How it gets smarter

Tunes to your payers and patterns; KG extended with your carrier rules (config, not code); confidence calibrated to your tolerance. **Improves from month 3+.**

“We can run a 48-hour proof-of-concept on a sample of your actual intake packages — no integration required, no PHI shared.”

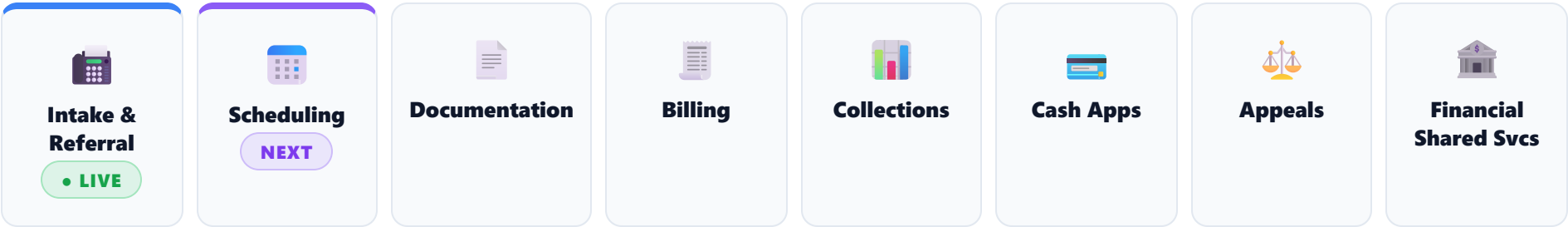
8–12 weeks from kickoff to go-live

A staged, low-risk path — shadow-mode pilot before anything goes live.



A platform, not a point solution

Referral to payment — an 8-agent value chain. Agent 1 is live; each agent is independently deployable in your environment.



Start where the pain is

Intake is live and proven. Scheduling is the natural next agent — but your priority shapes the order.

One architecture, many agents

Each agent shares the same auditable, human-in-the-loop design — deploy them one at a time, de-risked.

Walk in with a demo. Walk out with a plan.

Before we leave the room, we agree:

- **Pilot scope** — which referral types, which volume, shadow-mode start
- **Success metrics** — % auto-processed, accuracy, handling time, cost/referral
- **Owners & dates** — named owner and date next to every action
- **Security review** — HIPAA/BAA kicked off the week of June 16